ITT-370: Wireless Network Design Document

NETWORK DESIGN PROPOSAL

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Video Presentation: https://youtu.be/bomebXDqCsk?si=kQ_QIveCwJnSNwq6

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1. PROJECT SCOPE – Part 1

Project Charter:

The purpose of this project is to deploy a WLAN that supports two main digital advancements that the hospital requires for the efficiency, safety, and betterment of their healthcare staff, specifically nurses and doctors from every specialty and department. The hospital staff currently uses hard copies of medical records and paperwork, or has to review patient medical information manually in the nearest computer system prior to providing care. Replacing the proprietary standard wireless network will provide the speed, flexibility, security, and signal strength to provide nurses with an application and wireless device that will make sure that PHI is updated immediately and can be scanned via barcode to view patient data. In addition, doctors will be able to carry a IP phone while on duty so they can provide medical assistance without delay regardless of their location within the hospital building. The goal is to provide the hospital with as much ease and service to better the care they provide their patients and to provide the best environment for the healthcare workers. These tasks can benefit all healthcare providers but the budget is limited, and the allotted time and delicacy transitioning networks will be without hindering the reliance all medical applications, devices and services are utilizing 24/7.

Assumptions: The hospital assumes that there will be a positive outcome for the betterment of the healthcare providers and the patients thus increasing patient intake and revenue. It is assumed that all features must be aesthetically simple, easy to read, access, and utilize and must take into account the physical layout of the hospital with the various devices that may interfere with the needed network services. Liela will prepare a feasibility study as part of the project planning process and indicate the overall cost that is acceptable to the upper management of the hospital before funding and deployment can go forward.

Constraints: There are no current restrictions on hardware and software but there will be a restriction on budget that the hospital can allocate towards replacing its WLAN alongside its digital improvement sub-projects. This project cannot disrupt any of the applications, devices, and services that medical staff is relying upon to care for patients, which is forcing this transition to be a smooth, seamless and more gradual timeline. The information systems team need to provide their perspective and expertise on the matter and the new compliances that come with the 802.11 standards.

2. FEASIBILITY STUDY – Part 1

Objectives of the Network:

1. **Secure Service** –The main objective of this network is to provide an efficient, fast, reliable, and secure WLAN to provide the health care staff with the ability to host wireless devices with quick data delivery in a wide range of locations and to have IP phones available for all doctors on staff.
2. **Integration and Updates** – A seamless transition from the proprietary standard to the 802.11 standard wireless network while maintaining accessibility to healthcare staff in the hospital.
3. **Versatile Information Process** – The network needs to be able to be accessible in all relevant locations of the hospital, so that the IP phones can reach all doctors on duty, PHI is instantly updated and accessible via computer applications or wireless handheld devices. It must be able to host numerous computers and laptops for the nurses stations on each floor and in each doctor’s offices so they can access the same relevant applications as well. It must also host relevant medical devices that require network connection to return important PHI results or data to the healthcare providers who are monitoring it.
4. **Collaboration** – It will allow for seamless collaboration across multiple healthcare workers working together on the same patients, multiple shift changes, contacting different staff members and doctors when needed without delay so that medical treatment can be provided as soon as possible. It can increase the communication between patient needs/requests and the staff.

5. **Scalability** –Since the new network will be transitioned to 802.11, it is scalable and flexible enough to accommodate future needs and updates as new standards are developed in relation to the new digital requirements.

Intended Users – All relevant medical equipment/monitors, movable computers, IP phones, and handheld patient record scanning devices need to have a wide geographical signal coverage in order for nurses, CNA's, doctors and other healthcare staff can move in accommodation to patients or fellow staff members who need to visit different floors and departments. It must take into account the various departments that have equipment that will interfere with the network signal (MRIs, lead-containing equipment, X-rays, etc.).

Basic Equipment Needs and Proposed Budget- The proposed budget will be dependent on the full evaluation of square footage that needs to be covered, the specific amount of equipment and devices that need access to the services.

3. NETWORK NEEDS ANALYSIS – Part 2

Data Types – The type of data that will be used includes the following but not limited to the following application software:

- Electronic patient record software (EPR)
- Electronic medical record software (EMR)
- Medical database software
- Medical research software-
- Medical diagnosis software
- Medical imaging software-
- E-prescribing software- electronic prescriptions
- Telemedicine software
- Appointment scheduling (booking) (sign in) software
- Medical billing software
- Hospital management software
- Medical equipment management software
- Health tracking apps
- Personal Health Record software (medical diaries)
- Remote Patient Monitoring (RPM)
- Mobile health (mHealth) apps
- IP phone while on duty so they can provide medical assistance without delay regardless of their location within the hospital building

Data Sources – *How will the data be used?* Depending on the data and application, it will be stored in a database that will connect to other relevant databases/hospitals. In addition, it will be shared amongst relevant healthcare professionals when requested and will have the ability to be secure yet accessible so that if a patient needs a prescription, service or care it can be edited on their record and submitted as soon as possible. It will also need to be accessible by authorized personnel who seek to authenticate patient identity and details.

Number of Users that Will Need to Be Supported- Based on hospitals in my local area this hospital's users total to be about 3,500 (including physicians, nurses, technicians, cleaners, catering staff, contractors, patients, etc.)

Speed Requirements – *What type of expectations do you think the users will have for the network? What would be the throughput to meet these expectations?*

Due to the immediate nature in order to provide proper care to patients and to assist fellow healthcare staff, the need for instant/quick feedback, recommendations, updates, and assistance is a high priority. Due to legacy systems there will need to be a hybrid implementation to still integrate the 2.4 Ghz and add the 5Ghz bands.

Load Variation Estimates – *How much load do you think the network will be using?*

The load will remain steady but will fluctuate on the amount of patients and patient care that will be occurring at once. There will be subnets for different departments of care to alleviate the amount of traffic on the network. But with the amount of staff members and minimal outside users that will be utilizing the network for medical devices, applications, and the VoIP phones.

4. QoS REQUIREMENTS – Part 4

QoS Requirements – QoS requirements for phones, video, etc.

IP wireless phones require specific standards in order to operate officially without voice degradation, and to deliver telephonic data immediately.
The standards include having a consistent and quick jitter (consistent delay for each packet that is delivered) of less than 30 ms, delay in delivery time/latency from mouth to ear should be no more than 150 ms. A guarantee service (GS) will be implemented as it guarantees that packets will arrive within the guaranteed delivery time and not be discarded. One of the methods to ensure that IP wireless phones and other devices that require access to applications will be to segregate traffic on voice VLAN: 5 GHz band, because it is a larger variety of individual channels that can be utilized. Video data have similar requirements for delivery such as having a loss less than 1% and one-way latency no more than 150 ms.

From medical devices to workstations that access patient records devices need to have the ability to access application databases in order to provide updated, efficient, and accurate patient care as well as communicate any necessary details and updates to other medical staff/teams, insurance providers, etc. If devices fail to connect to the application database, they will not be able to send and/or receive accurate patient, prescription, insurance or other necessary information.
If any device in the hospital loses connection to application databases, patients may not be able to check in, receive care, receive prescriptions, access the most recent updates of needed care/surgery/procedures, and more. In addition, the hospital network will require QoS standards that are reliable, have timely delivery and guarantee of no packets being lost or delayed (hard guarantees).

With the high amount of traffic that will need to be delivered with a hard guarantee, there needs to be scheduling methods to ensure that all packets are treated fairly. The best method for the traffic will be priority queuing because it is time sensitive and will process the high priority packets first; there is risk of starvation for lower priority packets, but if we develop a methodology that allows the lower priority to be delivered on a different channel.

5. RELIABILITY AND SECURITY REQUIREMENTS – Part 5

Define Security Threats – Address all security threats.

Wireless networks are susceptible to a great amount of threats due to its ability to share services/data transmissions beyond most physical barriers. Some of the key threats include but are not limited to the following:

- Energy Jamming- A malicious actor intentionally attempts to interfere/prevent/reduce with the reception, communication, and transmission of data by interjecting electromagnetic waves on the same frequency that the reader uses.
- DOS aka Denial of Service- There is a wide variety of methods to achieve a DOS attack; it is when the attacker floods the network and/or communication between devices with so much content or traffic to stop or hinder its intended services.
- Random Access Channel (RACH) flooding is when an attacker tramsits a huge amount of access request over the RACH to hinder the ability of legitimate users to access the gNB (gNodeB) which is a node in a cellular network that provides connections to the evolved packet core
- Access Management Protocol Exploitation can be caused due to unauthorized access to the IAM and/or not having reliable authentication.
- Rogue access points- When a malicious or innocent actor tries to attach a non authorized access point to a network (usually to boost service or track information), this can create a series of network vulnerabilities. It will not be properly assimilated with all of the security protocols and softwares or it can give a malicious actor full or alot of access to the network.

Security Measures and Configuration – Describe the effectiveness/ineffectiveness of these measures and configurations.

Reactive Jamming Detection using Bit Error Rate (BER) is one of many of the proposed security proactive ways to combat energy jamming attacks. There are many that have been tried and proven to be ineffective such as channel hopping, but there have been false negatives due to miscalculation of the bit errors. Due to the wide variety of physical and technical factors that play a role in energy jamming; it is hard to settle on one tried and true method to rely on, security protocols will require reflecting on past evidence, in-depth knowledge of the used channels and physical architecture.

DDOS comes from a variety of starting point and has many different variations which is why it is important to have a proactive security architecture and protocol in place. Ensuring that the IAM is implementing strong password policies, keeping software and hardware patched and up to date, ensuring that users are not being captivated by phishing attacks or downloading insecure/irrelevant attachments from outside sources. The hospital will have a DDOS Attack Threat Model that the security team will adhere to as a reference to identifying suspicious activity on the network on each relevant layer (databases, Network Layer, Transport Layer) and ensure that there is a set list of priorities of services within the network.

Having a well developed and reliable IAM is necessary in order to remain compliant with HIPAA and other PHI privacy protections. Poor password hygiene, poor management of user permissions or logging of events can cause anomalies to go under the radar and for unauthorized access, manipulation and use to occur. In addition, we recommend that since most of the laptops and workstations will be using Microsoft/Windows OS that Kerberos will be the authorization tool.

To help reduce the temptation of adding rogue access points and to provide resistance for malicious actors who want to add their own devices to a network there are some basic security protocols that can be implemented. For instance having hardware setup so that it is not easy to install an outside hardware without it resisting a foreign or unapproved device. Second, ensure that hardware is being monitored by cameras at all times, has security sensors so that IDS teams can see when or if a device is being compromised , and are able to locate where that is physically.

Encryption Methods – Best encryption methods you can deploy on a wireless network to protect user authentication and data.

Since the hospital is going to be utilizing 802.11 and needs the best performance of encryption while data is in transit due to its sensitive nature, WPA2 encryption methods will be utilized. It is compatible with 802.11n and 802.11ac networks.

To provide secure transmission within the WSN of the biosensors to connect patient vital status to the diagnostics centre and other PHI related data, symmetric algorithms and attribute-based encryption is needed to secure data transmissions and the access control systems for the medical sensors network connections. A key encryption technique that will be utilized is the blowfish algorithm which is a secure block cipher that is faster and is symmetric.

Risk of Employees Taking Laptops Home – If there are hospital staff that are approved to bring their work laptop home with them, there needs to be a strong Work From Home Technology Security Policy that users will need to read and agree to.

With the hospital working with extremely private records the staff will need to understand that these matters are not to be printed at home, easily accessible or displayed at public places, and the laptop will have all the necessary security develops such as patched/updated by logging into the company's network at least once a week, mandated hospital provided VPN, refrain from using free Public hotspots unless it is proven to be legitimate, having management systems ensure that all users are abiding by the strong password policies, mandate that data is backed up at least once a week if relevant to the user.

Federal Guidelines – According to the government requirements, what type of security will need to be put in place?

In a hospital or any healthcare setting, the majority of the data at rest, in transit or in use is some form of healthcare information and/or personal health information (PHI). This means that all data that is moving throughout the hospital's network can not be obtained by unauthorized parties, manipulated or accessed without authorization by the patient or a legal representative of the patient. In Healthcare, there are four main laws that require cybersecurity proactively and action in response to the dramatic increase in malicious activity against healthcare (data).

HIPAA- Health Insurance and Accountability Act of 1996 published the Privacy Rule and Security Rule which est. standards for protecting specific health information and specific health information that is being held or transferred in electronic form/ e-PHI.

OSHA-Occupational Safety and Health Act of 1970 forced manufacturers to report on safety incidents and accidents which applies to the high cybersecurity risks that industrial institutions face.

PCI- The Payment Card Industry developed a data security standard for cardholder data that all organizations must comply with when processings, storing or transmitting credit card data.

6. DETAILED DESIGN DOCUMENTATION – Final Project

Aprox. 1 million square feet with over 3500 users + additional medical equipment/devices requiring services

Note: List of Equipment needed:

- ***1000 Access Points:***
    - ***Cisco Catalyst 9130AX Series Access Point- $1300***
    - ***Total Cost: 1,300,000***
    - ***Availability: Available/In Stock***
    - ***Performance:***
        - 800-1000 sq ft (800 to clients and 1000 to other APs)
        - 8x8 uplink/downlink MU-MIMO with eight spatial streams
        - Uplink/downlink OFDMA
        - TWT

        - BSS coloring
        - MRC
        - 802.11ax beamforming
        - 20-, 40-, 80-, and 160-MHz channels
        - PHY data rates up to 5.38 Gbps (8x8 80 MHz or Dual 4x4 80+80 MHz on 5GHz and 4x4 20 MHz on 2.4)
        - Packet aggregation: A-MPDU (transmit and receive), A-MSDU (transmit and receive)
        - 802.11 DFS
        - CSD support
        - WPA3 support
    - ***Maintainability:***
        - Operating temperature: 32° to 122°F (0° to 50°C)
        - Operating humidity: 10% to 90% (non condensing)
        - Operating altitude test: 40˚C, 9843 ft (3000 m)
        - Note: When the ambient operating temperature exceeds 40°C, the access point will shift from 8x8 to 4x4 on the 5 GHz radio, uplink Ethernet will downgrade to 1 Gigabit Ethernet; however, the USB interface will remain enabled.
        - Easily scalable
- ***Zoom Phone-Business Plus***
    - ***Cost: $269.90/year/user or 2699.00 annually***
    - ***Available: Subscription***
    - Meetings up to 30 hours per meeting
    - 300 Attendees per meeting
    - Whiteboard
    - Team Chat
    - Mail & Calendar
    - Client & Service
    - Clips Plus
    - Notes
    - AI Companion
    - Cloud Storage 10GB
    - Essential Apps
    - Free premium apps for 1 year
    - Extras SSO, managed domains & more
    - Scheduler
    - Phone Unlimited regional
    - Translated Captions
    - Workspace Reservation
- ***2 Cisco VG350 Analog Gateways***
    - ***Cost: $9000 each***
    - ***Total Cost: $18000***
    - ***Available: Available/In Stock***
    - Standalone solution for high-density for deployments of up to 160 analog ports.

    - Provides mix and match modules for high density long or short loop lengths solutions
    - Increase flexibility with additional slots available for FXS and FXO ports
    - Improve end-user experience using advanced digital signal processor technology to reduce background noise and prevent acoustic shock
    - Advance power management using Cisco EnergyWise technology
    - Supporting traditional analog devices on IP networks. These devices include analog phones, fax machines, modems, voicemail systems, and speakerphones.
- ***Cisco Catalyst 9300 Series Switches***
    - **Cost**: $2000 each
    - Encrypted Traffic Analytics (ETA)
    - Plug and Play (PnP) enabled
    - Support for AES-256 with the powerful MACsec 256-bit encryption algorithm
    - Policy-based automation from edge to cloud.
    - Dual-stack support for IPv4/IPv6 and dynamic hardware forwarding table allocations, for ease of IPv4-to-IPv6 migration.
    - **Availability: Available/In Stock**
- **Print Z-Band ZD510-HC Printers**
    - Easiest wristband printer
    - Smartchip in each cartridge identifies wristband size and automatically configures darkness/speed settings
    - Adhesive closure
    - Resistance to hand sanitizers
    - Softest Direct Thermal Wristband
    - **Price**: $622.45 each
    - **Quantity:** 50
    - **Availability**: Available/In Stock
- **Zebra Medical Barcode Scanner- DS8178-HC**
    - cordless scanner for healthcare environments
    - 800 MHz microprocessor
    - highest sensor resolution
    - high density focus
    - PRZM Intelligent Imaging Technology
    - Built-In Lamp
    - Eliminate Bluetooth with WiFi Friendly Mode
    - **Price**: $300 each
    - **Available**: Available/In Stock
    - **Quantity**: 1250

7. COST ANALYSIS – Final Project

Cost Analysis			
Tangible Costs			
Product	**Price**	**Quantity**	**Total**
Print Z-Band ZD510-HC Printers	$622.45	75	***$46683.75***
Zebra Medical Barcode Scanner-DS8178-HC	$300 each	1250	***$375000***
Cisco Catalyst 9130AX Series Access Point	$1300	1000	***$1,300,000***
Zoom Phone-Business Plus	2699.00 annually		***$2699***
Cisco VG350 Analog Gateways	$9000	200	***$1,800,000***
Cisco Catalyst 9300 Series Switches	$2000	500	***$1,000,000***
Subtotal of Costs			***$4,524,382.75***